

JAYANTH DASAMANTHARAO

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PROFESSIONAL EXPERIENCE

AI Engineer - Fedway Associates, EliteUS Software Solutions Aug 2024 - Present | Edison, NJ

- Deployed production GenAI and ML-serving systems supporting real-time customer workflows, leveraging Python-based backend services, embeddings, RAG with a focus on reliability and low-latency inference.
- Improved model quality and service reliability by implementing automated evaluation, regression detection, A/B testing, and production monitoring for online inference systems.
- Built scalable data and retrieval pipelines including document chunking, hybrid semantic search, reranking, and vector-based storage, supporting online inference for thousands of users with consistent performance.
- Developed and maintained backend ML services and APIs using Python and FastAPI, with logging, authentication, guardrails. Resolved latency and production issues to maintain system health and uptime.

Data Science Intern, Harvest Software Solutions June 2023 - Sept 2023 | Jacksonville, Florida

- Engineered scalable ML pipelines and monitoring systems for large advertising datasets, increasing campaign visibility from 15% to 45% while ensuring reliability and performance.
- Built ML services and NLP APIs using XGBoost, H2O AutoML, and FastAPI, improving sentiment classification accuracy by 65% and campaign spend efficiency by 50%.
- Designed SQL/Python data pipelines and model evaluation workflows, applying core data structures and algorithms to support efficient feature generation and experimentation.

Application Development Associate, Accenture June 2021 - Aug 2022 | Hyderabad, India

- Developed and optimized Python-based ETL and data processing pipelines using hash maps, sets, indexed lookups to efficiently clean, deduplicate, and transform large datasets, reducing production issues by 50%.
- Implemented SQL workflows and joins with partitioning, indexing, and window functions, enabling efficient aggregation and analysis over millions of records in AWS Athena and Redshift.
- Applied batching, caching, and algorithmic optimizations for data validation and pipeline execution, improving system performance and reliability by 25%.

TECHNICAL SKILLS

Programming:	Python, R, SQL. Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, FastAPI.
Machine Learning :	Model Fine-tuning, evaluation and experimentation, deployment, Feature Engineering.
GenAI & LLMs:	LLM fine-tuning, RAG, embeddings, prompt optimization, LangChain, multimodal modeling.
AI Platforms/Agents:	ML serving systems, GenAI platforms, memory architectures, tool calling, guardrails.
APIs & Services:	REST APIs, API versioning, authentication, developer tooling.
Cloud & MLOps:	PostgreSQL, Data Warehousing, AWS (Bedrock, Athena, Redshift), GCP, Azure, Vectordb.

EDUCATION

Rutgers University, Master of Science in Data Science Sept 2022 - May 2024 | New Brunswick, NJ

Personal Website: <https://jayanthdasamantharao.github.io/>

Andhra University B.Tech in Electrical and Electronics Engineering 2017 - 2021 | Visakhapatnam, India

RESEARCH PROJECTS

Self-Driving Cars (Computer Vision): Built autonomous driving cars with vision pipeline using CNNs, YOLO, Faster R-CNN, achieving 95%+ object detection accuracy.

Multimodal ML Serving System (Production): Built and deployed backend services for retrieval and generation using embeddings and multimodal inputs, focusing on reliability, latency, and production robustness.

Bayesian Classification: Implemented Bayesian logistic regression in PyMC3 to predict diabetes status, incorporating informative priors, achieving 0.85 F1-score.

For additional projects and code samples: <https://github.com/JayanthDasamantharao>